

Building User Interfaces

Voice User Interface Design & Experience Prototyping

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What will we learn today?

- Introduction to Conversational Interface
- Design principles for VUI
- Experience prototyping

Introduction to Conversational Interface Technologies

What is a conversational interface technology?

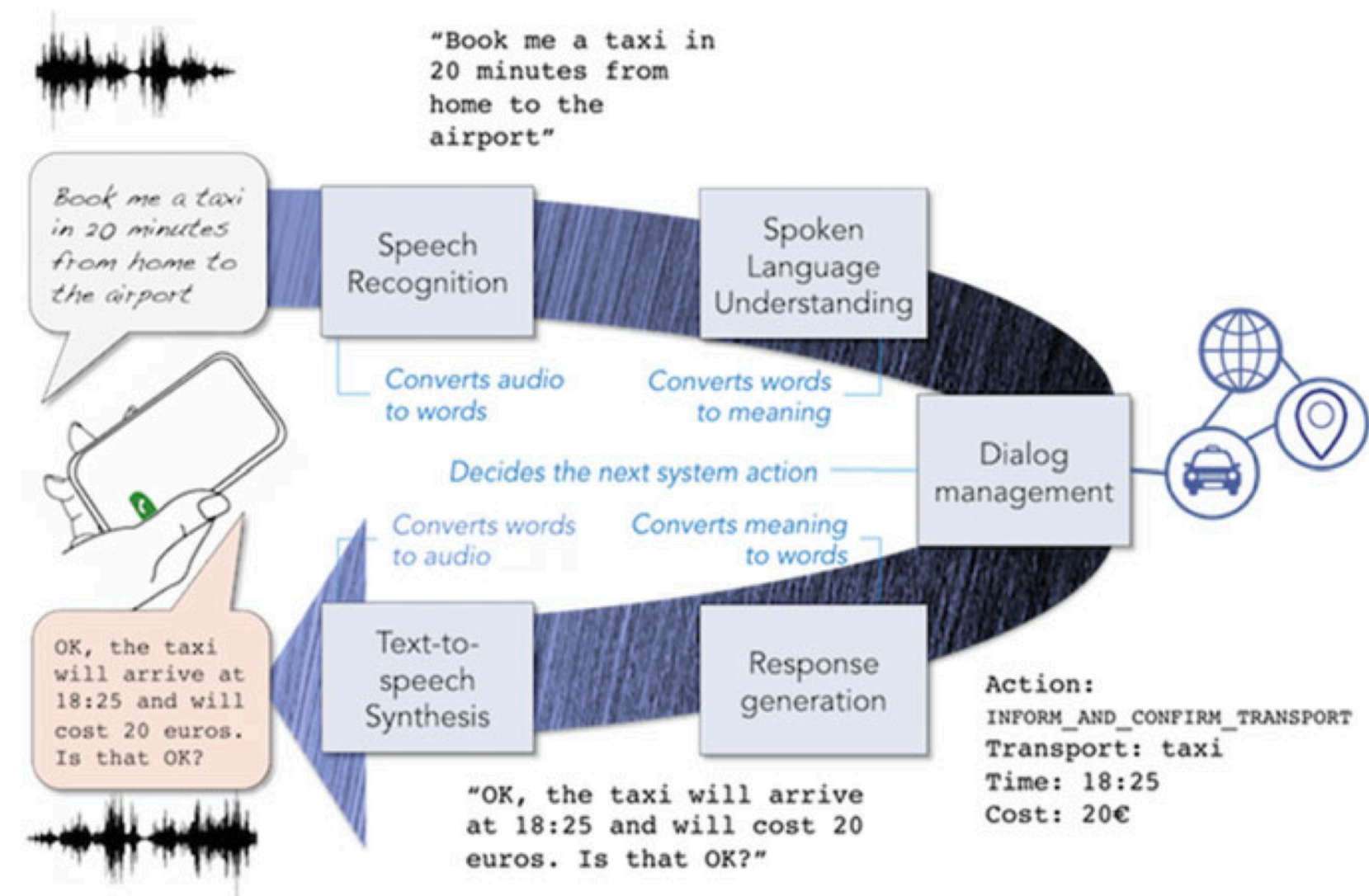
Definition: "Technology that supports conversational interaction with these VPAs [virtual personal assistants] by means of speech and other modalities."¹

¹ McTear et al., 2016, *The Conversational Interface*

Components of Conversational Interfaces²

Conversational interfaces integrate a number of technologies:

- Speech recognition
- Spoken language understanding
- Dialog management
- Response generation
- Text-to-speech synthesis



² Image source: McTear et al., 2016, *The Conversational Interface*

Platforms for Building Conversational Interfaces

In this course, we will be using Dialogflow by Google.

However, there are alternatives:

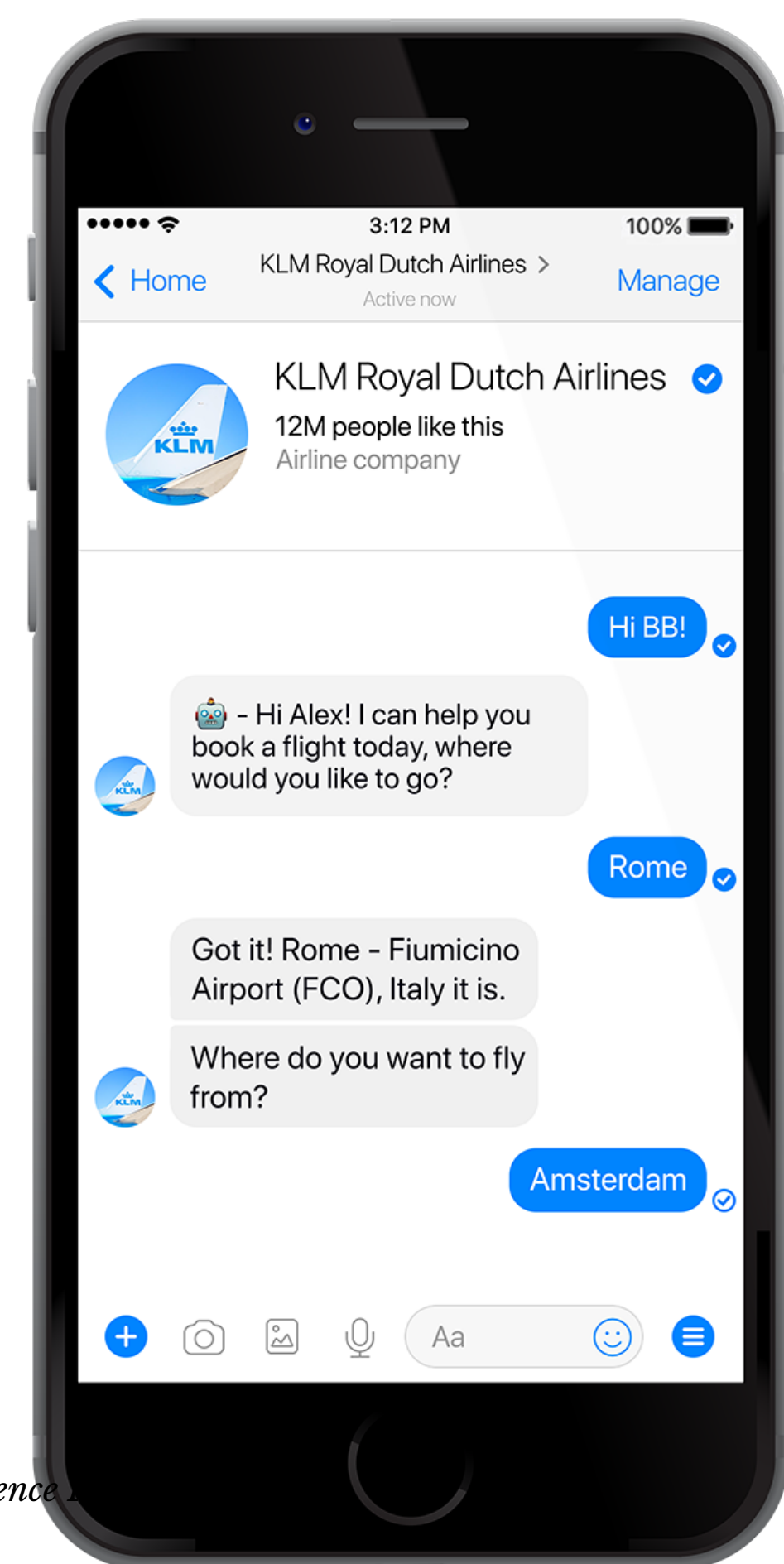
- Amazon Lex (Commercial)
- Mycroft (Open Source)
- CoreNLP (Academic)

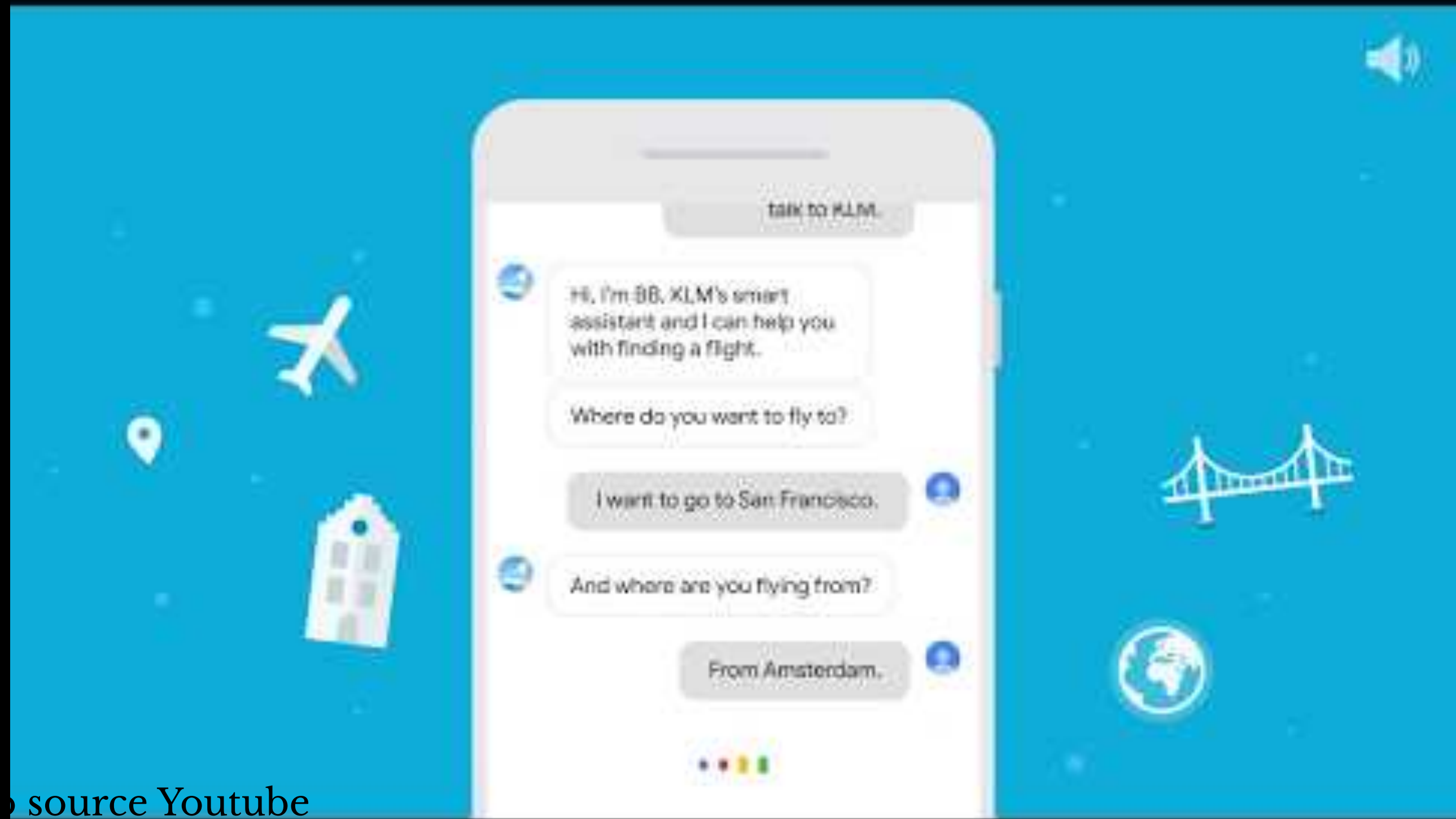
Case Study: KLM BB³

KLM used Dialogflow to create an agent to purchase travel as well as travel preparation.

Let's see how the KLM BB works...

³ Image source [Dialogflow](#)





source Youtube



source

Design Principles for Conversational Interfaces

Elephant in the Room

Recap: Definition of Usability: The effectiveness, efficiency, and satisfaction with which a specified set of users can achieve a specified set of tasks in a particular environment. — ISO 9241-11

Conversational interfaces are almost always less effective, less efficient, and less satisfactory than graphical user interfaces.

Effectiveness: Conversational interfaces are more error prone due to technology, ambiguities, and environmental influences.

Efficiency: Using conversational interfaces is almost never as fast as using graphical user interfaces.

Satisfaction: Interacting with conversational interfaces can be awkward, socially inappropriate, and frustrating.

So, what is the point of conversational interfaces?

Where do these interfaces deliver value?

1. Streamlining app installation, login, payment, notifications, and and so on in a conversational paradigm.⁶
2. In some contexts, e.g., while driving, CIs are more effective, efficient, and satisfactory due to resource constraints.
3. CIs address many accessibility problems, including vision (e.g., blindness), motor (e.g., tremor), and cognitive (e.g., dyslexia) deficiencies.

⁶ Grover, 2016, Bots won't replace apps. Better apps will replace apps.

Design Principles

Gricean Maxims⁷

Definition: Proposed by Paul Grice, conversations follow the cooperative principle and four key maxims:

- Maxim of quality (truthful and accurate communication)
- Maxim of quantity (just the right amount of information)
- Maxim of relevance (appropriate and relevant information)
- Maxim of manner (clear, cooperative communication)

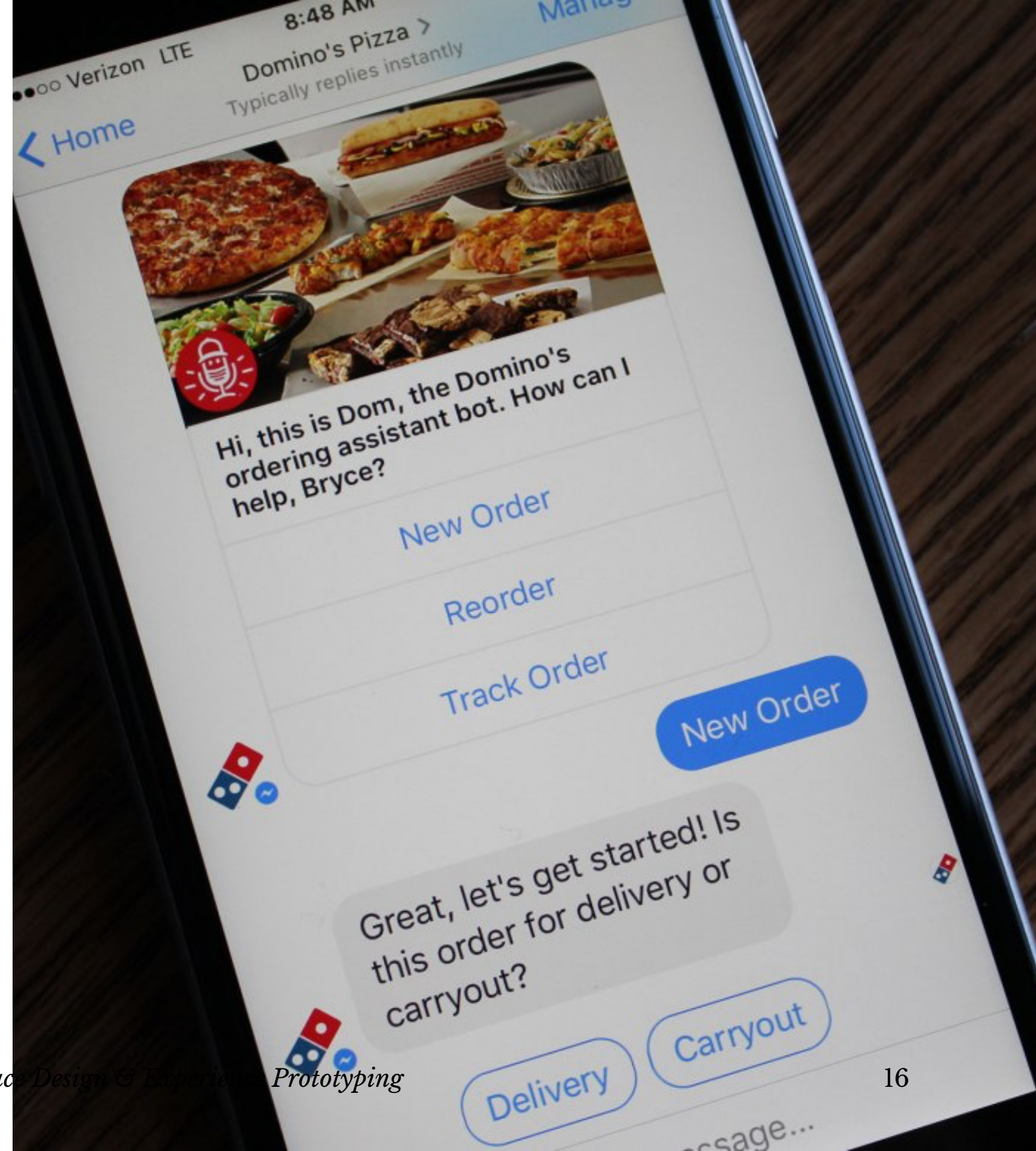
⁷ Grice, 1975, *Logic and Conversation*

Multimodality⁸

Definition: Multimodal interfaces utilize multiple modalities, including visual information, speech, touch, and so on, in user experiences they afford.

Most conversational interfaces are multimodal interfaces.

⁸Image source



Multimodality Principle: Take advantage of other modalities, e.g., visual information, vibrations, etc., wherever appropriate.

Using multimodal components, you can provide users with breaks for decision making, interruptions, etc.

Potential caveats:

- Ask, "does my interface still support a speech-only interaction?"
- The conversational and other components must be designed together to fit within the conversation.

Interaction Paradigm

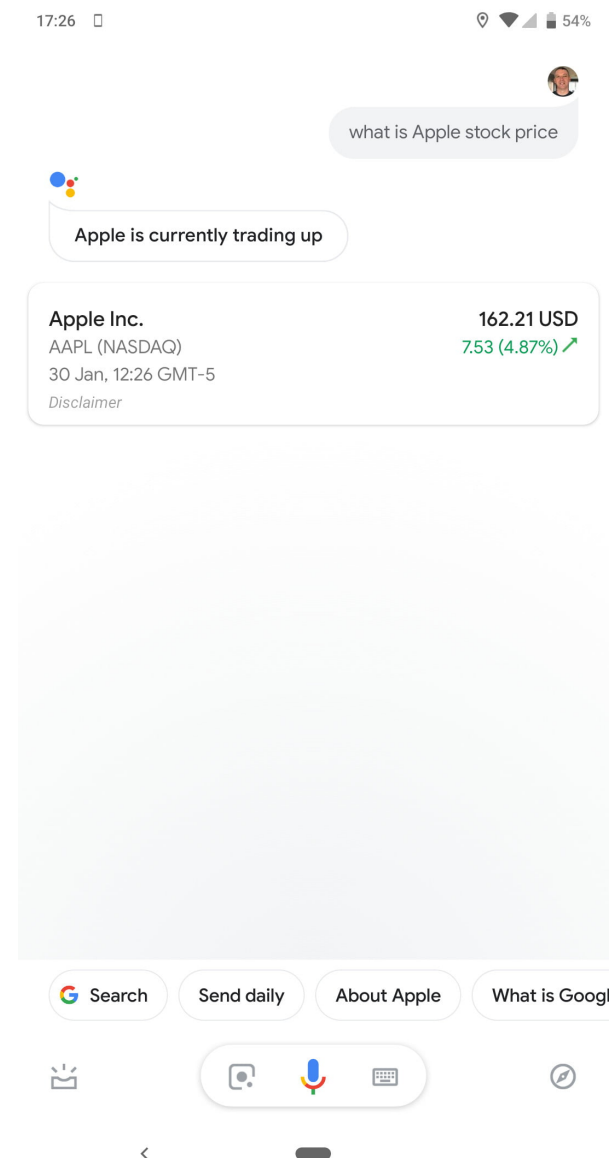
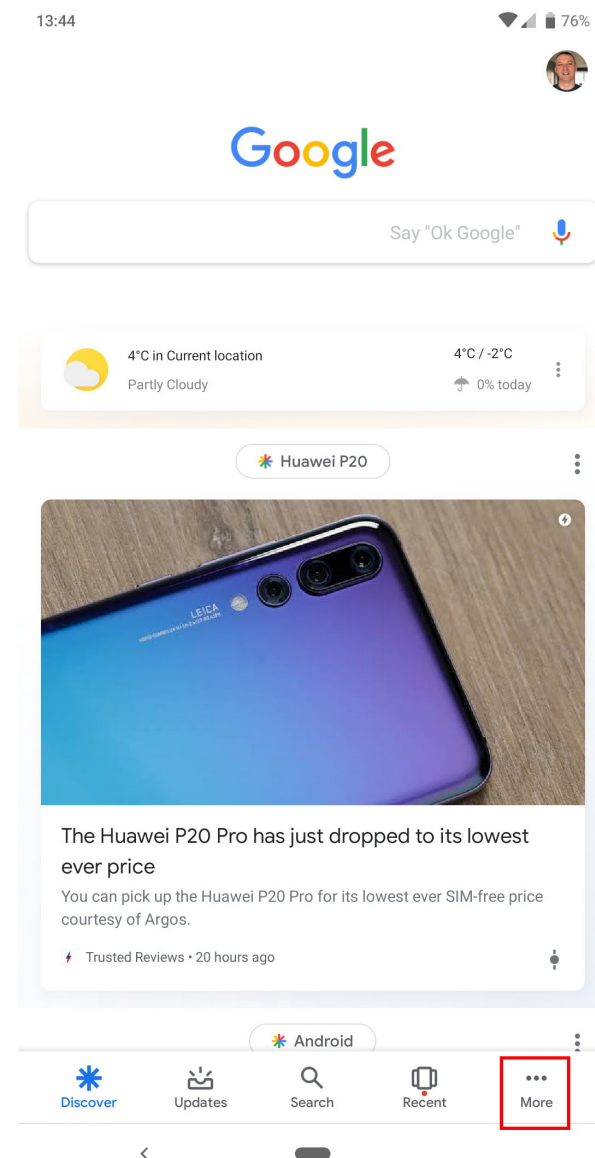
Conversational interfaces can follow different paradigms depending on the context of use and the design of the application:

- Command-and-control interfaces (always-on voice assistants)
- Conversational interfaces (chatbots, task assistants, social robots)

Command-and-Control Interfaces

Definition: Interfaces where speech input is mapped to specific system functions that are called immediately. These interfaces commonly utilize:

1. Expressing user intent using a wake word (e.g., "OK, Google") or the pressing of a button (e.g., home button in the iPhone)
2. Indicating listening and understanding
3. Executing the mapped function



⁹ Image source: left, right, middle

Conversational Interfaces

Definition: Interfaces where the interaction with the system has the characteristics of human conversations, including turn taking, theoretically infinite depth, conversational markers, etc.

Turn-taking

Definition: Speaking turns are the core, cooperative structure of conversations that involves one speaker at a time and an explicit exchange of tokens.

Principles:

1. One speaker at a time — *transparency in who is speaking*
2. Turn exchanges — *explicit signaling of who will speak next*
3. Interruption handling — *very difficult with CIs*

Conversational Markers¹⁰

Definition: Speech cues that indicate the state or the direction of the conversation. Types of conversational markers:

1. Timelines ("First," "Halfway there," "Finally")
2. Acknowledgements ("Thanks," "Got it," "Alright," "Sorry about that")
3. Positive feedback ("Good job," "Nice to hear that")

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Assistant: I'll be asking you a few questions about your health. First, how many hours of sleep did you get last night?

User: About seven.

Assistant: Good job. And how many serving of fruits and vegetables did you eat yesterday?

User: Maybe four.¹⁰

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Assistant: Got it. Last question—were you able to take your medication last night?

User: Yes.

Assistant: All right. That's it for now. I'll talk to you again tomorrow. Goodbye.¹⁰

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Confirmations¹⁰

Definitions: CIs are designed with explicit forms of confirmation to improve system usability and transparency.

Can be explicit vs. implicit and speech-based vs. non-speech based (visual, action).

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Explicit confirmation: Requiring the user to confirm: "I think you want to set a reminder to 'buy insurance before going skydiving next week.' Is that right?"¹⁰

Implicit confirmation: Letting user know what was understood: "Ok, setting a reminder to buy insurance..."

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Error Handling

Definitions: Deviations from expected conversational flow due to technical mistakes, unexpected user behavior, environmental influences, etc.

Types of errors:¹⁰

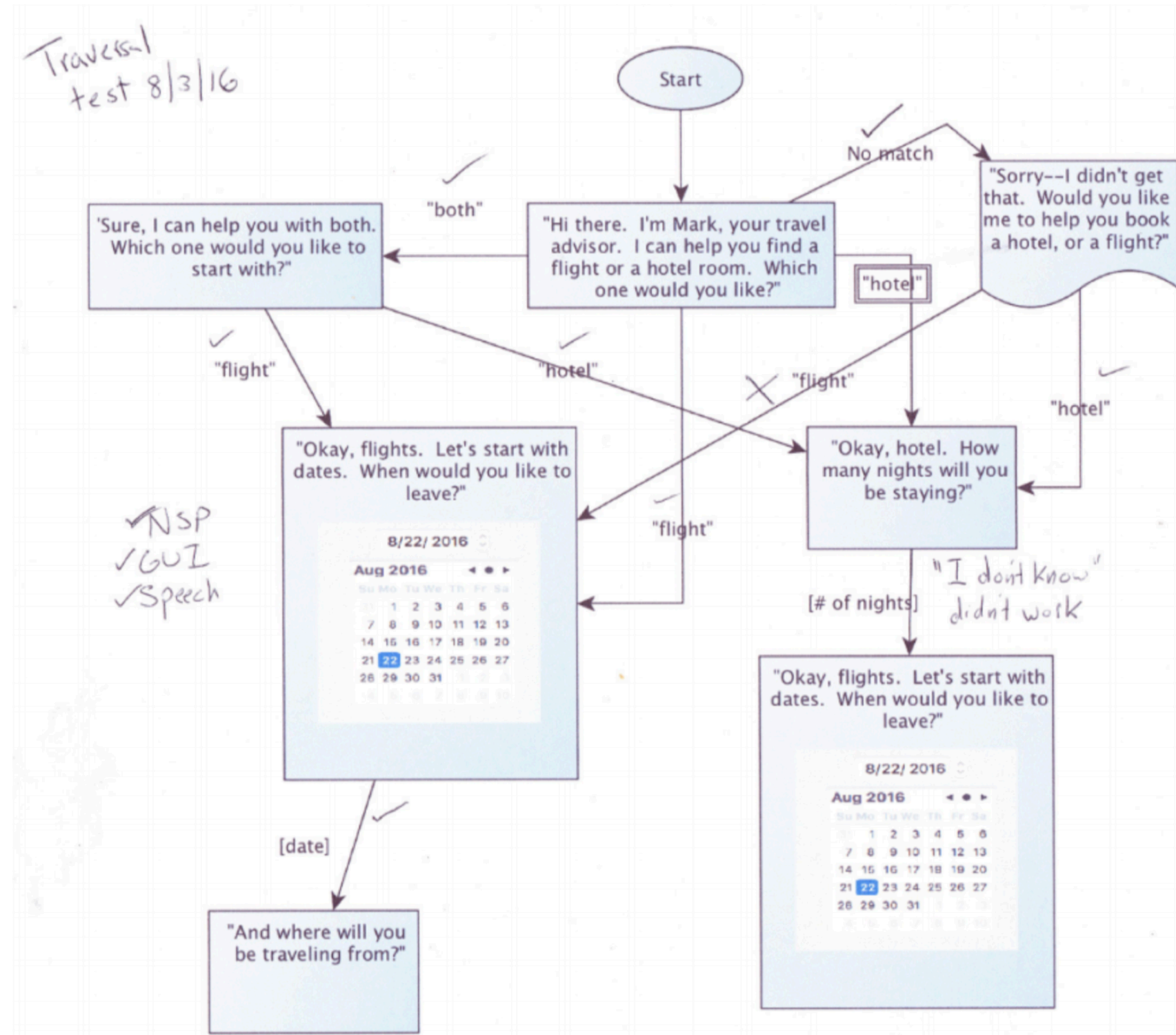
1. No speech detected
2. Speech detected, but nothing recognized
3. Something was recognized correctly, but the system does the wrong thing with it
4. Something was recognized incorrectly

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences

Flowcharting Conversational Interactions¹⁰

The most commonly used method of modeling and prototyping conversational interactions is defining flows that show how the interaction will flow depending on system state, user behavior, or external influences.

¹⁰ Pearl, 2016, Designing Voice User Interfaces: Principles of Conversational Experiences



Heuristics for Conversational Interfaces¹¹

Seventeen heuristics that fall into five broad categories:

1. General
2. Conversational style
3. Guiding, Teaching, and Offering Help
4. Feedback and prompts
5. Errors

¹¹ Wei & Landay, 2018, Speech-based Conversational Agent Heuristics

Experience Prototyping

The Problem¹²

Conventional prototyping methods provide limited support for *conversational* interfaces.



¹²Image Source

Why is Conversational Interfaces hard to prototype?

Human Dialogue

Social interactions are driven by *tacit knowledge*:¹³

...we can know more than we can tell...

An evolutionarily encoded and culturally situated set of rules, patterns, and practices for effective interpersonal communication.

How do we design interfaces that follow these rules, patterns, and practices?

¹³ Michael Polanyi, 1958, *Personal Knowledge*

The Solution: Experience Prototyping

Definition: Prototyping the holistic experience of interacting with a product.

A related definition:¹⁴

An experience prototype is any kind of representation, in any medium, that is designed to understand, explore or communicate what it might be like to engage with the product, space, or system we are designing.

¹⁴ Buchenu & Fulton Suri, 2000, *Experience Prototyping*

Wait, what?¹⁵

How does experience prototyping solve the problem of designing interfaces to follow human norms of interaction?

We have tacit knowledge about how conversational interactions work. By *acting out* an interaction, we apply our knowledge to a scenario.

What I hear I forget. What I see, I remember. What I **do, I understand!⁸**

¹⁵ Chinese proverb

⁸ Image source

We use ourselves as *decoders* for the norms encoded in us!¹⁶



¹⁶ Image source

When do we do experience prototyping?

Three key uses:

1. Understanding existing user experiences and context
2. Exploring and evaluating design ideas
3. Communicating ideas to an audience

What is it that we prototype?

1. System behavior
2. User behavior
3. Interactions with context

How do we do experience prototyping?

Step 1: Define context¹⁷

What is the context of the interaction?

E.g., passengers using entertainment system on a bus, travelers packing their luggage.



¹⁷ Image source

Step 2: Develop Scenarios¹⁸

What are concrete interaction scenarios do we want to support?

E.g., buying a ticket, users packing, cooking a meal.



¹⁸ Image source

Step 3: Identify Design Goals¹⁹

What role does my design play in these scenarios? How does it support the user in the target activity? What capabilities will it offer?

E.g., find, filter, and purchase flights; help the user set and follow personal goals through daily reminders.



¹⁹ Image source

Step 4: Set up the Environment²⁰

How can I represent the context of the interaction?

E.g., creating props to represent devices, environmental constraints.



²⁰ Image source

Step 5: Act out Interaction²¹

How will the interaction unfold? How will the user behave? How should the system behave?



²¹ Image source

Bodystorming²²

Definition: *Bodystorming* is a creativity method that involves physically experiencing a situation to develop new ideas and insights.



²² Image source

Step 6: Develop Insight²³

What did you learn about system behavior, user behavior, and interactions with context?



²³ Image source

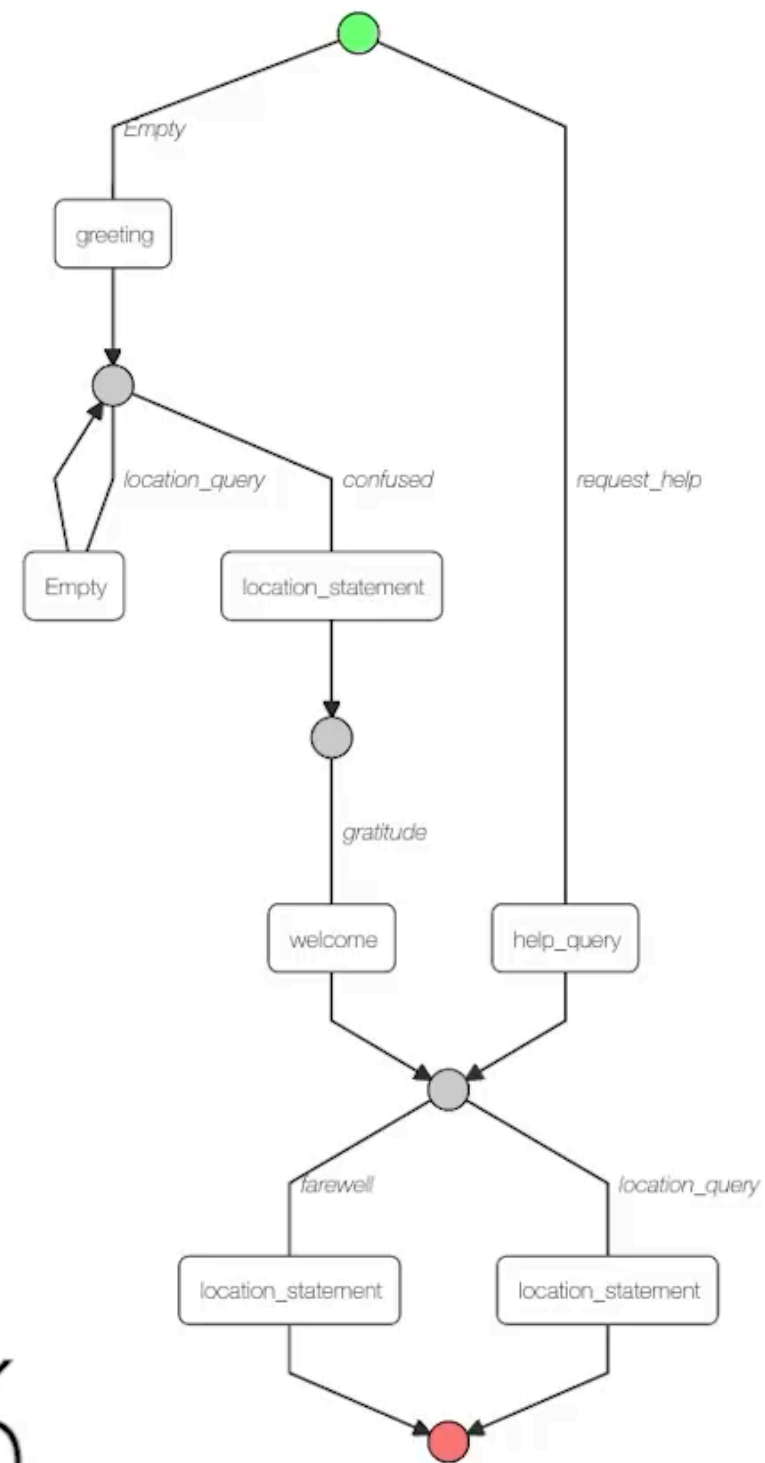
Pro Tip: Experience prototyping is *very* awkward. Get the awkwardness out of your system so that you can focus on using the method for design.

Example Use of Bodystorming²⁴

Supporting design teams in ideating and acting out human-robot interactions using a system called Synthé.

²⁴ Porfirio et al., 2019, *Bodystorming Human-Robot Interactions*

Synthé



Bodysessions

on/off	bodysession information		
☒	0. Session4	Length: 2	Sat.: 100%
☒	1. Session8	Length: 5	Sat.: 100%

Connection Status: ✓✓

Parameters

Maximum number of states

0 5 10

View Options

Visualization

☒ Model ☐ Traces

Font Size 12

Display nonverbal behaviors ☐

Additional resources

- Seminar paper on experience prototyping
- Case studies in bodystorming
- "Universal Methods of Design": sections on *experience prototyping* and *bodystorming*

What did we learn today?

- Intro to Conversational Interfaces
- VUI Design
- Experience Prototyping